

## 1 The Earth's atmosphere

Scientists use evidence from rocks and other sources to estimate how the composition of the Earth's atmosphere has changed over billions of years.

Fig. 1.1 shows estimates for the early atmosphere and the atmosphere today.



Fig. 1.1

| Gas            | Early atmosphere (%) | Modern atmosphere (%) |
|----------------|----------------------|-----------------------|
| carbon dioxide | 95.00                | 0.04                  |
| oxygen         | 0.00                 | 21.00                 |
| nitrogen       | 1.00                 | 78.00                 |

- (a) Calculate the decrease in the percentage of carbon dioxide between the early atmosphere and the modern atmosphere.

Give your answer as a percentage point decrease.

Answer: \_\_\_\_\_ percentage points [2]

- (b) Calculate the percentage decrease in carbon dioxide in the atmosphere. Use the equation below.

$$\text{percentage decrease} = (\text{original value} - \text{new value}) / \text{original value} \times 100$$

Give your answer to 3 significant figures.

Answer: \_\_\_\_\_ % [2]

- (c) The percentage of oxygen increased from approximately 0% to 21%. Explain how living organisms caused this change. [3]

---

---

---

**1 The Earth's atmosphere (continued)**

- (d) Carbon dioxide in the Earth's atmosphere decreased greatly over time. Explain two processes that caused the percentage of carbon dioxide in the atmosphere to decrease.

[4]

---

---

---

---

---

---

- (e) Nitrogen now makes up about 78% of the Earth's atmosphere. Suggest why nitrogen was able to build up in the atmosphere over time.

[2]

---

---

---

---

**Total: 13 marks**

# Mark Scheme

## 1 The Earth's atmosphere

Award marks using the guidance below. Accept equivalent scientifically correct wording unless the mark scheme states otherwise.

| Question | Answer / marking points                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks | Guidance                                                                                                                                                                                                      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)      | $95.00 - 0.04 = 94.96$ percentage points                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     | 1 mark for correct subtraction / working.<br>1 mark for 94.96 percentage points.                                                                                                                              |
| (b)      | $((95.00 - 0.04) / 95.00) \times 100 = 99.957\ldots\%$<br>Answer = 100% to 3 significant figures.                                                                                                                                                                                                                                                                                                                                                                                                   | 2     | 1 mark for correct substitution into the equation.<br>1 mark for the correct final answer. Accept 99.96% if the learner has not rounded to 3 s.f.                                                             |
| (c)      | Photosynthesis by early photosynthetic organisms such as algae, and later green plants, removed carbon dioxide and released oxygen (O <sub>2</sub> ).                                                                                                                                                                                                                                                                                                                                               | 3     | Award 1 mark each for: <ul style="list-style-type: none"> <li>• photosynthesis</li> <li>• algae / green plants / photosynthetic organisms</li> <li>• release of oxygen / removal of carbon dioxide</li> </ul> |
| (d)      | Any two processes, explained. For example:<br><br>Oceans and carbonate rocks: as the Earth cooled, water vapour condensed to form oceans. Carbon dioxide (CO <sub>2</sub> ) dissolved in the oceans and became locked into carbonate sediments / rocks such as limestone.<br><br>Photosynthesis and carbon storage: photosynthetic organisms removed CO <sub>2</sub> from the atmosphere. Carbon was then stored in biomass and, over geological time, in fossil fuels and carbon-containing rocks. | 4     | Award up to 2 marks for each correctly explained process.<br>Maximum 2 marks if only one process is described.<br>Accept other valid mechanisms, e.g. carbon locked in shells and skeletons.                  |
| (e)      | Nitrogen (N <sub>2</sub> ) is relatively unreactive, so once released it was not readily removed from the atmosphere and could gradually accumulate.                                                                                                                                                                                                                                                                                                                                                | 2     | 1 mark for low reactivity.<br>1 mark for the consequence: not readily removed / gradual build-up.                                                                                                             |

**Total: 13 marks**